

Introduction to ECMO Fundamentals

Let's start our discussion of the fundamentals of extracorporeal membrane oxygenation (ECMO) by returning to our graph. We have spent the last few chapters developing and unpacking the simple but essential concept that the amount of support that is needed for critically ill patients is associated with a dose-related toxicity that increases at higher levels.

The majority of the concept was developed in the context of optimizing oxygen delivery, but the same concept exists for any intervention that is performed in the intensive care unit (ICU). The more sedation, antibiotics, and medications that are needed to support the patient, the greater the potential for drug interactions as well as progressive weakness, delirium, and immobility. It is not simply that sick patients need more support – the support itself can lead to a progressive downward spiral with diminishing positive returns and progressive toxicity that can lead to decompensation and death.

Let's propose now that for specific patients, we can support them with ECMO, a modality that may be associated with some increased risk, but which can hypothetically decrease the degree of dose-related toxicity associated with conventional care (Fig. 6.1).

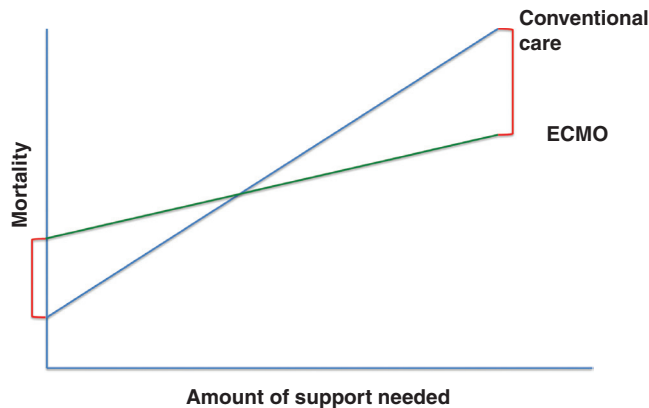


FIG. 6.1 The risk-benefit rationale of ECMO versus conventional care

SO WHAT IS ECMO?

Extracorporeal membrane oxygenation, or ECMO, is a form of extracorporeal life support that functionally involves a blood pump and membrane lung or oxygenator, to support the heart and/or lungs. Let's look at a basic schematic.

The mechanism is deceptively simple – pumping the blood out of the body, oxygenating the blood, and returning that blood back to the body (Fig. 6.2).

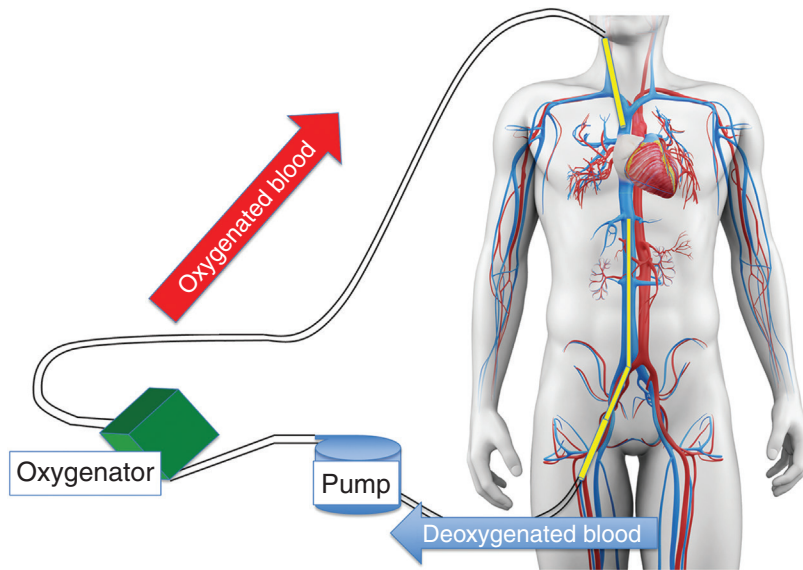


FIG. 6.2 ECMO circuit simplified (Modified from SciePro/Shutterstock.com)

As straightforward as that mechanism is, the physiology of the body that it interacts with can be complex, and forms the basis of the discussions to follow, to truly understand the concepts, physiology, and management principles behind ECMO.

WHY DOES THE ECMO CIRCUIT SEEM MORE COMPLICATED WHEN I LOOK AT IT?

Fig. 6.3 is an example of what might be seen in an ECMO circuit.

It is important to first understand the basics of what ECMO is trying to accomplish. From there, we can fill in the other components that are part of the ECMO circuit. ECMO circuits

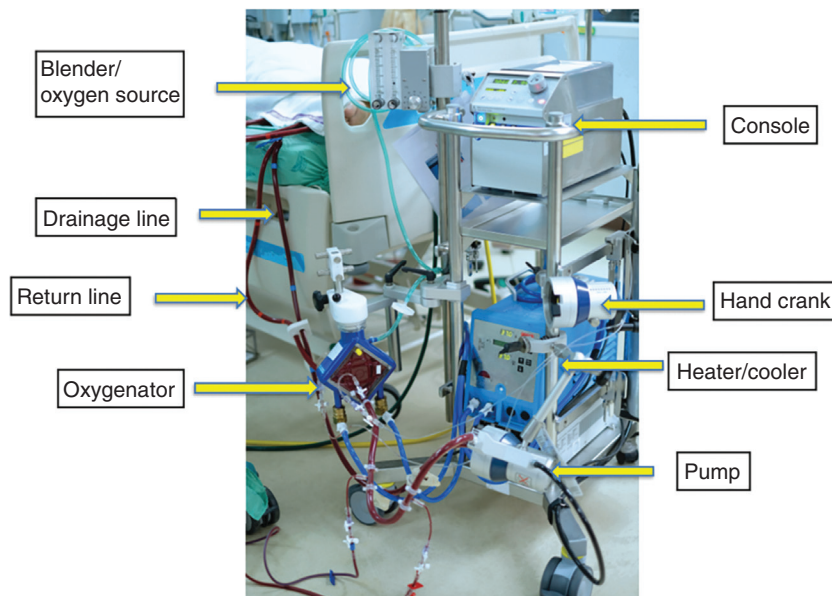


FIG. 6.3 ECMO circuit component (From PIJITRA PHOMKHAM/Shutterstock.com)

have become more straightforward in recent years, but the complexity of the circuit that you are managing can differ widely and is largely determined by institutional protocols, preferences, and processes.

Each component has serves a specific function. Often if this is difficult to distinguish, it can be helpful to trace the circuit in the direction of the blood, reminding ourselves that the basic design is drainage of deoxygenated blood through a pump, through a membrane oxygenator, with the return of oxygenated blood back to the patient.

Let's now better familiarize ourselves with all of the other components, using Fig. 6.4 to help discover various parts of the circuit. We will start in the direction of blood proceeding from the drainage cannula, through the circuit, and finally to the return cannula.

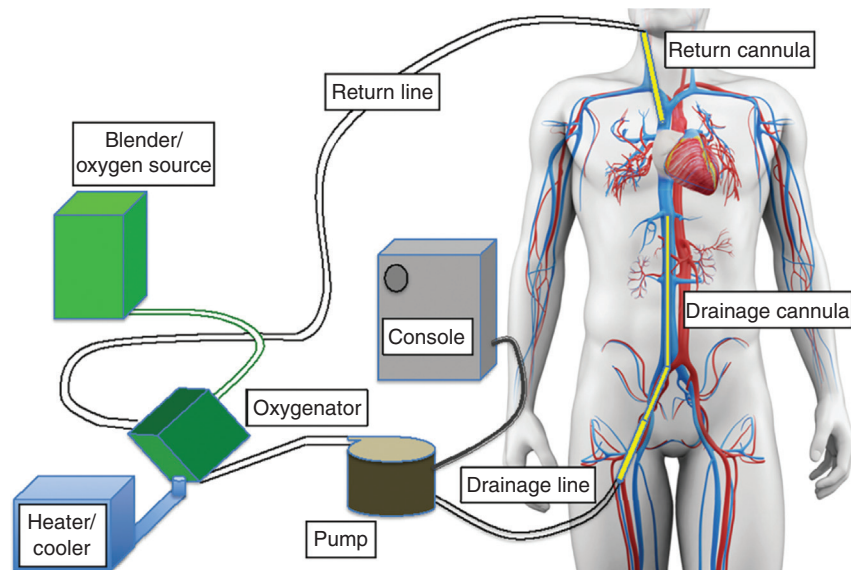


FIG. 6.4 Schematic of ECMO circuit components (Modified from SciePro/Shutterstock.com)

Drainage Cannula

The drainage cannula is the entry point into the ECMO circuit. It is a large-bore access tube that is implanted into the venous circulation through a variety of techniques, including surgical through a graft sewn onto the blood vessel, percutaneously, or through cutdown. It can usually be distinguished from the return cannula in that it is draining dark, deoxygenated blood (Fig. 6.4).

Drainage Line

This is usually a set of clear plastic tubing that is connected to the drainage cannula. The convention is for drainage tubing to be 3/8 of an inch, but this can differ. The drainage line is usually distinguished by the presence of blue tape; however, if this tape is not present, then the color of the blood can be revealing (Fig. 6.4).

Pump

The pump is probably one of the most important components of the ECMO circuit – indeed there is no extracorporeal membrane oxygenation without the ability to circulate blood extracorporeally! However, despite this relative importance of the pump, it can be difficult to notice at first.

If you do not know what you are looking for, then you may miss it. Note the pump, where the blood is entering and where it is exiting. Also at this point, note the backup for the pump, which should always be within arms reach, whether a hand crank, backup console, or backup circuit altogether (Fig. 6.4).

Console

The console is like the dashboard of your car – even though it has nothing to do with actually moving the vehicle forward, it is the interface that you are most likely to interact with. Some consoles can be very straightforward, denoting the speed of the pump and the rate of blood flow only, while others can have a lot of bells and whistles with other data points. The console will usually have some mechanism for altering the revolutions per minute (RPMs), whether a dial or push button (Fig. 6.4).

Oxygenator

The oxygenator is where the membrane oxygenation of ECMO occurs. There are many different types of oxygenator, the common thread being that they have a mechanism for oxygenating the blood through diffusion, and usually a mechanism for temperature regulation (Fig. 6.4).

Blender/Oxygen Source

The oxygenator will be connected to some source of oxygen, usually in the form of a blender, which can be mixed with medical grade air to allow for differing concentrations of oxygen to be delivered. The blender will have the ability to alter the FiO_2 of the gas delivered as well as the rate of the gas flow (known as sweep gas flow). In cases of transport or in emergencies, this oxygen connection can take the form of an oxygen tank, which can deliver 100% FiO_2 at differing sweep gas flow rates (Fig. 6.4).

Heater/Cooler

As we will cover later, the oxygenator is formed by the interface of three phases – blood, oxygen/sweep gas, and water. The water is used to temperature regulate the blood that is going back to the patient, either warming it to account for ambient loss of temperature in the circuit or cooling it in the case of fever or targeted hypothermia (Fig. 6.4).

Return Line

The return line looks almost identical to the drainage line, with the notable exception that it is filled with oxygenated/bright red blood. The line may also be denoted with red tape, which can be especially important when being connected as a new circuit (Fig. 6.4).

Return Cannula

The destination of the return line is a second cannula that is inserted into the venous or arterial circulation, depending on the type of support being offered. In some cases, the return line can be connected to the return port of a dual lumen cannula that returns blood. We will go into much more detail on the types of drainage/return cannulas when we discuss configurations in Chapter 8.

Overall, that's it! Those are the primary components common to almost every ECMO circuit that you will encounter. There are other components, many of which we will cover in more detail in Chapter 9, but these are the common components that achieve the primary objective of ECMO – drain blood, oxygenate/remove CO_2 , temperature regulate, and return the blood to the body. Keep the overall flow straight and you will have a systematic approach to any circuit that you come into contact with.

Now let's go into more detail about what is happening with the fundamental component of ECMO – the oxygenator.

UNDERSTANDING THE OXYGENATOR: THE FUNDAMENTAL MECHANISM OF ECMO

The essence of ECMO is returning oxygenated blood to the body. The oxygenator is where all of this happens. When blood flows through the oxygenator, it enters through an inlet and then follows a specific path, in order to allow for blood to easily pass through while minimizing the entrainment of any air that may come through the circuit (Fig. 6.5).

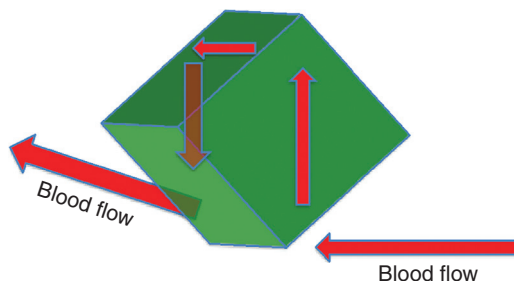


FIG. 6.5 Blood flow through the oxygenator

The oxygenator itself is made of a collection of tiny, hollow plastic tubes that allow for oxygen to flow through. As the blood flows through the oxygenator, it diffuses around the tubes counter-current to the direction of the flow of gas (Fig. 6.6).

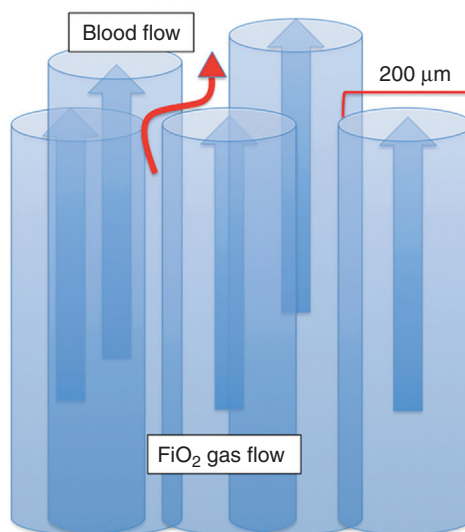


FIG. 6.6 Blood flow through the fibers of the oxygenator

At this level, there is a blood-gas interface, much in the same fashion as what happens at the alveolar level. Just like the alveolar interface, this blood-gas interface allows oxygen to flow down its concentration gradient, from the gas phase in the fibers to the blood, while also allowing CO_2 to go down its concentration gradient, from the blood back into the gas (Fig. 6.7).

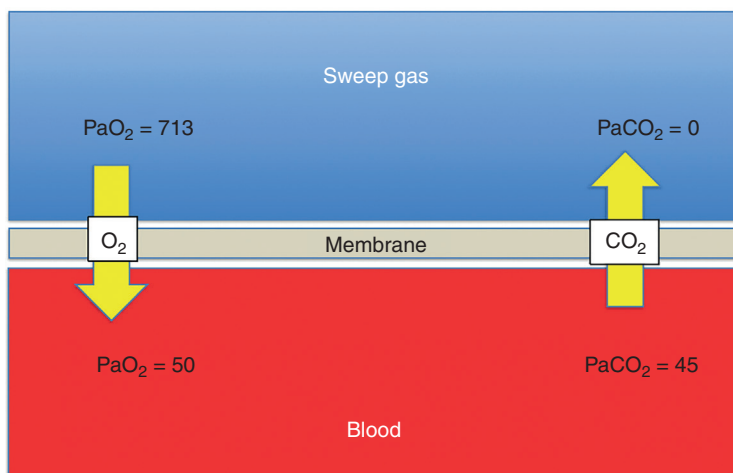


FIG. 6.7 Diffusion of oxygen and carbon dioxide at the membrane level

While all of this is occurring, temperature-controlled water is diffusing around the blood and gas tubules. The water is in a completely separate system – it doesn't come into contact with the blood or sweep gas, but only functions to warm/cool the blood through conduction (Fig. 6.8).

The result is that as blood leaves the oxygenator, it is now oxygenated and temperature controlled, with CO_2 removed. This blood can then be returned to the body.

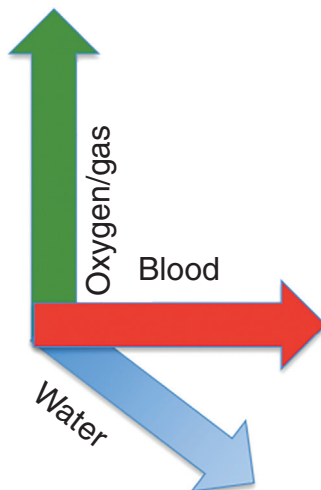


FIG. 6.8 The blood-sweep gas-water interface within the oxygenator

WHERE DOES THE BLOOD GO FROM HERE?

There are two distinct types of ECMO that are defined by where the oxygenated blood is returned. If the blood is drained from the venous side, oxygenated, and returned back into the venous system, it is referred to as veno-venous ECMO, VV ECMO, or sometimes simply respiratory ECMO (Fig. 6.9).

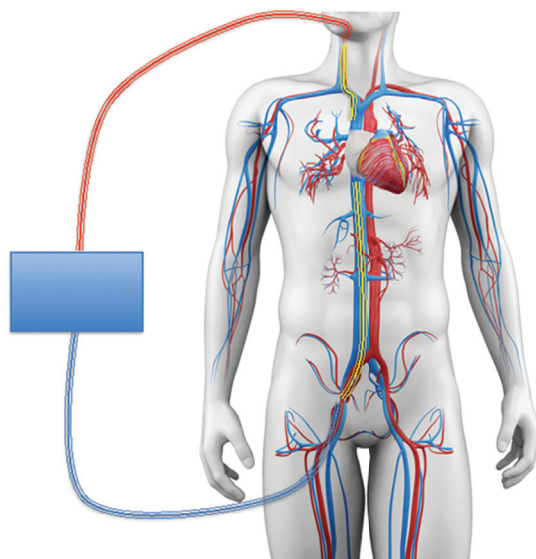


FIG. 6.9 Configurations: veno-venous (VV) (Modified from SciePro/Shutterstock.com)

This configuration allows for oxygenated blood to be returned to the right side of the heart but is dependent on the heart to pump the blood through the circulation. If a significant shunt exists, say due to acute respiratory distress syndrome (ARDS) or pneumonia, the effect of the shunt is reduced because the shunted blood is better oxygenated. If you were not on ECMO, the shunted blood would be shunting from the venous system across the lungs and to the rest of the body.

In contrast, if we were to send the oxygenated blood into the *arterial* system, we now have veno-arterial ECMO, or VA ECMO, or sometimes simply cardiac ECMO (Fig. 6.10).

This configuration allows for oxygenated blood to be returned to the arterial side, so that some/all of the blood is completely bypassing the heart. Any blood that runs through the circuit,

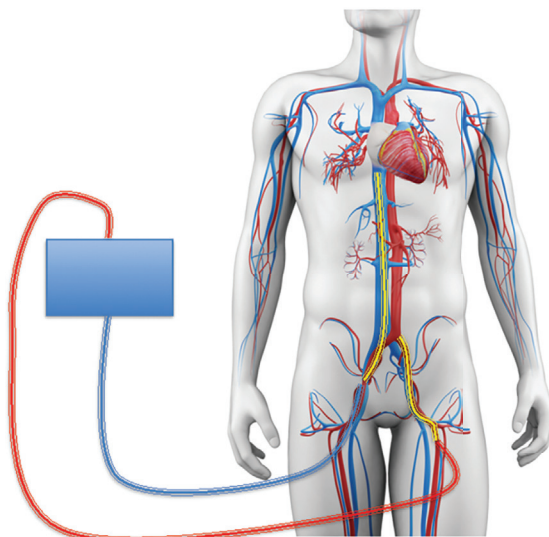


FIG. 6.10 Configurations: veno-arterial (VA) (Modified from SciePro/Shutterstock.com)

therefore, contributes to the perfusion of the body and is *not* dependent on the native cardiac output to do so.

There are many physiologic implications of these two types of support that we will cover in the chapters to come, but for now in the most introductory sense, these are the two primary ways that oxygenated blood can be returned to the body.

HOW CAN I MANIPULATE THE CIRCUIT?

There are really only a couple of mechanisms that can be manipulated on an ECMO machine:

1. RPMs/blood flow
 2. Sweep gas flow
 3. Temperature
- That's it. Let's review.

Revolutions Per Minute

Altering the RPM is one of the most fundamental changes that can be done to the circuit. Increasing the RPMs increases the speed that the centrifugal pump turns and, consequently, can induce higher blood flow. There are many factors that affect how RPMs correlate with flow, which we will cover in greater detail when we discuss flow physiology in [Chapter 10](#).

For the present time, suffice it to say that an increase in RPMs is analogous to a motor in a car – the faster the motor turns, the faster the car goes.

RPMs are usually manipulated on the console, whether in the form of a dial or a button. There will also be a display on the console that will denote the number of RPMs that the pump is currently running at as well as the consequent blood flow ([Fig. 6.11](#)).

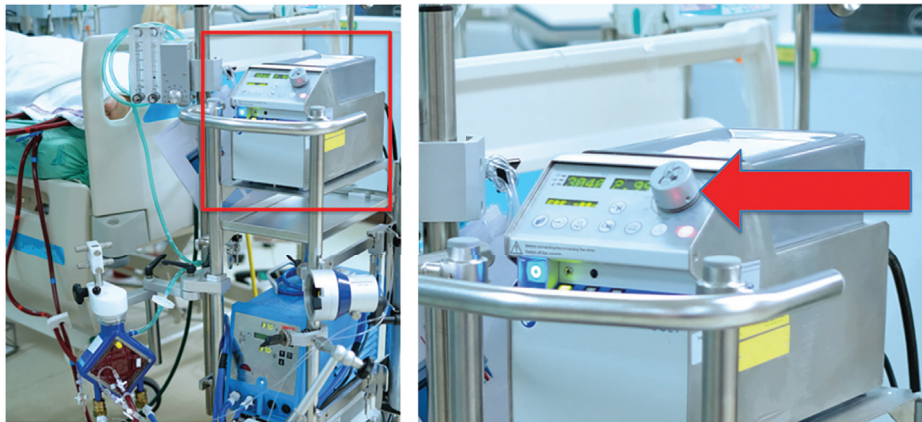


FIG. 6.11 ECMO settings: revolutions per minute/flow (From PIJITRA PHOMKHAM/Shutterstock.com)

Sweep Gas Flow/FiO₂

Sweep gas is the gas that is infused into the membrane. There are two fundamental changes that can be made to it – the rate of gas flow and the FiO₂ of this gas ([Fig. 6.12](#)).

Usually the manipulations are made to the blender as denoted earlier. In this case, the knob on the far right allows for changes in FiO₂, from 21% (room air) to 100% FiO₂.

The sweep gas flow rates are adjusted with the knob to the left of the FiO₂ knob, going from 0 to 10 L/min. The knob to the far left is for smaller increments of the final liter of flow from

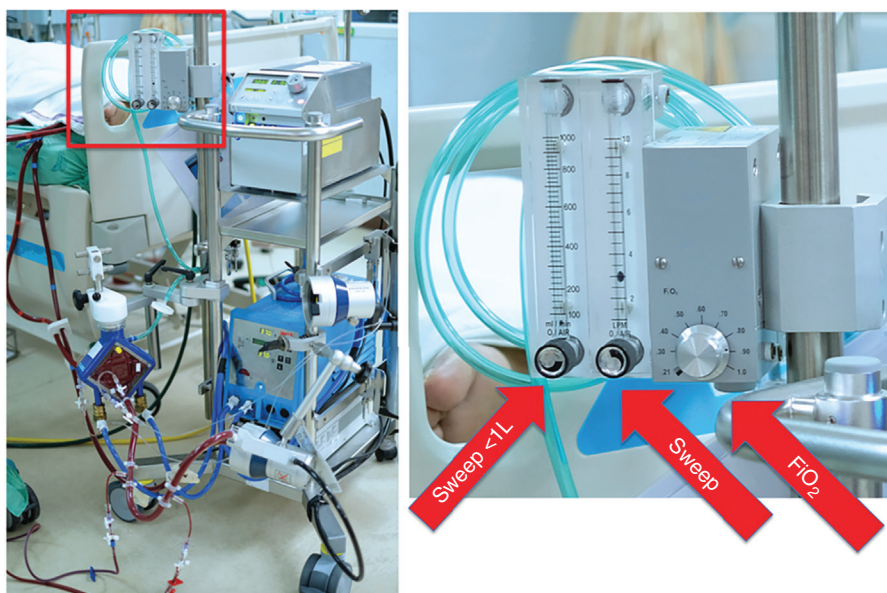


FIG. 6.12 ECMO settings: sweep gas flow/ FiO_2 (From PIJITRA PHOMKHAM/Shutterstock.com)

0 to 1000 mL (1 L) per minute. Note that any flow on this knob is in addition to the flow demarcated in the middle knob.

What Are the Effects of Manipulations to Sweep Gas Flow and ECMO Blood Flow to the Patient?

We will go into this in much more detail in the ECMO Physiology section, but for now, the basic direction is as follows:

FiO_2 and ECMO blood flow affect **oxygenation/oxygen delivery** while sweep gas flow affects **ventilation/ CO_2 removal**

Temperature

The final setting that can be adjusted on the ECMO circuit is the temperature of the blood that is returned to the body. This adjustment is made much less frequently than blood flow or sweep gas flow/ FiO_2 . That said, this mechanism allows for the close maintenance of temperature, even to the tenth of a degree (Fig. 6.13).

PUTTING IT ALL TOGETHER

We went over a lot in this chapter in order to give a very high-level overview of ECMO – to include the rationale behind the use of ECMO, how the ECMO circuit works, introduction to the components, configurations, and, finally, how the circuit can be manipulated.

Hopefully you will agree that the rationale and mechanism behind ECMO is relatively straightforward. This does not mean the management of ECMO itself is straightforward. Instead, as we will continue to appreciate, the complexity arises from the physiology with which the ECMO circuit interacts. However, we will continue to develop a systematic approach to the

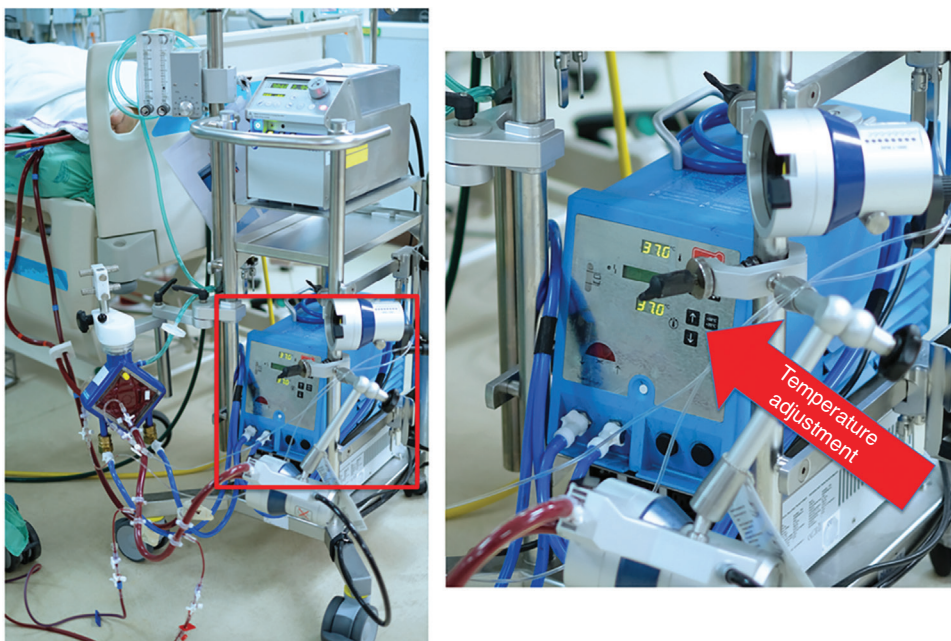


FIG. 6.13 ECMO settings: temperature (From PIJITRA PHOMKHAM/Shutterstock.com)

various aspects of the ECMO circuit, including its physiology, in order to build confidence and understanding of the concepts related to ECMO.

But we are just getting started. We will continue to develop this approach in the next chapter when we take on how we can approach selecting patients for ECMO.

SUGGESTED READING

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